CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 24-2025: A 32-Year-Old Woman with Fatigue and Myalgias

Deborah Gomez Kwolek, M.D.,[1,2] Julian S. Haimovich, M.D.,[1,2]
Marc D. Succi, M.D.,[3,4] David M. Dudzinski, M.D.,[1,2]
and Sarah E. Turbett, M.D.[1,2,5,6]

PRESENTATION OF CASE

Dr. Julian S. Haimovich: A 32-year-old woman was evaluated at this hospital because of fatigue and myalgias.

Two and a half years before the current presentation, the patient had fatigue, headache, myalgias, and “brain fog” after infection with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Evaluation of these symptoms included consultations with neurology, immunology, and rheumatology specialists at another hospital. The blood levels of electrolytes, thyrotropin, alanine aminotransferase, aspartate aminotransferase, bilirubin, alkaline phosphatase, C-reactive protein, and ferritin were normal, as were the complete blood count and differential count and the results of tests of kidney function. Serologic tests for cytomegalovirus, Epstein–Barr virus, ehrlichia species, anaplasma species, and *Borrelia burgdorferi* were reportedly negative. The results of magnetic resonance imaging of the head were reportedly normal.

During the next 2 years, the patient received acupuncture therapy and used herbal supplements, which helped to alleviate her symptoms. Five weeks before the current presentation, the patient was reinfected with SARS-CoV-2. After 5 days, symptoms associated with acute SARS-CoV-2 infection abated and she returned to her usual state of health.

Nine days before the current presentation, the patient began to have neck tightness and pain after lifting heavy objects. The pain radiated to the head and scapulae. Six days before the current presentation, she was evaluated by her primary care physician, who was affiliated with the other hospital. Examination was reportedly normal. Treatment with ibuprofen, magnesium, acupuncture, and massage were recommended.

The neck pain did not abate. Three days before the current presentation, the pain began to radiate down the right arm, and severe fatigue developed. The patient presented to the emergency department of a second hospital for evaluation. On examination, the temporal temperature was 36.8°C, the heart rate 50 beats per minute, and

Author affiliations are listed at the end of the article.

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CME

the blood pressure 109/55 mm Hg. She had full range of motion in the neck and tenderness on palpation of the right side of the upper neck and the right side of the upper back. A radiograph of the cervical spine was reportedly normal. Treatment with intravenous ketorolac, oral methocarbamol, and transdermal lidocaine was administered, and a tapering dose of oral methylprednisolone was prescribed.

During the next 2 days, the back and neck pain resolved. However, the fatigue worsened and myalgias developed. The patient had palpitations and an irregular pulse, as well as waxing and waning lower chest pain that she described as tightness. The chest pain was both pleuritic and positional in nature, and she rated the pain at 5 on a scale of 0 to 10 (with 10 indicating the most severe pain). She presented to the emergency department of a third hospital for evaluation.

On examination, the temporal temperature was 36.3°C, the heart rate 58 beats per minute, the blood pressure 124/79 mm Hg, and the oxygen saturation 100% while the patient was breathing ambient air. The blood levels of electrolytes and troponin I were normal, as were the complete blood count and the results of tests of kidney function. A urine test for human chorionic gonadotropin was negative. The blood level of alanine aminotransferase was 170 U per liter (reference range, 7 to 40), the aspartate aminotransferase level 104 U per liter (reference range, 8 to 30), and the D-dimer level 1760 ng per milliliter (reference value, <520). Electrocardiography (ECG) showed first-degree atrioventricular block and a heart rate of 55 beats per minute. The results of chest radiography and an ultrasound examination of the liver were reportedly normal. Additional imaging studies were obtained.

Dr. Marc D. Succi: Computed tomographic (CT) angiography of the chest, performed after the administration of intravenous contrast material that was timed for evaluation of the pulmonary arteries, showed no evidence of filling defects that would suggest pulmonary embolism. The heart and lungs appeared normal. There was no evidence of mediastinal or hilar lymphadenopathy. The bones had a normal appearance.

Dr. Haimovich: The patient was discharged with a plan for the use of mobile cardiac telemetry. The next day, the chest pain worsened and dyspnea on exertion developed. The patient also had dizziness, bifrontal headache, and chills. Her primary care physician recommended that she seek evaluation in the emergency department of this hospital.

In the emergency department, the patient reported ongoing fatigue, myalgias, and chest tightness. She had no fever, rhinorrhea, pharyngitis, paresthesia, weakness, or diplopia. A review of systems was notable for intermittent abdominal discomfort, nausea, and diarrhea that had been present for several months.

Other medical history included umbilical hernia repair, hyperemesis gravidarum, anxiety, and mild chronic gastritis. Medications included magnesium supplementation, probiotics, and a supplement containing turmeric, animal liver extract, and milk thistle. The patient had not been vaccinated against SARS-CoV-2. There were no known adverse reactions to medications.

The patient was married and lived with her husband and two children in a forested town in a neighboring state. Rabbits and sheep were raised on her property. She was physically active, participating regularly in hiking, camping, and yoga. She rarely consumed alcohol. She vaped and had a history of tobacco and marijuana use. She did not use other substances. The patient's family history was notable for hypertension, a patent foramen ovale, and stroke in her maternal grandmother; hypertension in her maternal grandfather; colorectal cancer in her paternal grandmother; and Kawasaki's disease in one of her children.

On examination, the temporal temperature was 36.3°C, the heart rate 52 beats per minute, the blood pressure 140/76 mm Hg, and the oxygen saturation 100% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 19.6. The cardiac rhythm was irregularly irregular. The remainder of the examination was normal.

The blood levels of electrolytes, calcium, magnesium, thyrotropin, albumin, bilirubin, alkaline phosphatase, C-reactive protein, creatine kinase, and high-sensitivity troponin T were normal, as were the complete blood count and differential count, the erythrocyte sedimentation rate, and the results of tests of kidney function. The blood level of alanine aminotransferase was 89 U per liter (reference range, 7 to 33), the aspartate aminotransferase level 22 U per liter (reference range, 9 to 32), the D-dimer level 436 ng per milliliter (reference value, <500), and the N-terminal pro–B-type natriuretic peptide (NT-proBNP) level 604 pg per milliliter (reference value, <450). Imaging studies were obtained.

Dr. Succi: A single-view frontal chest radiograph obtained with a portable device (Fig. 1) showed normal lungs without evidence of pneumonia, pulmonary edema, or pleural effusion. The heart size was normal, and there was no radiographic evidence of mediastinal lymphadenopathy. The bones had a normal appearance.

Dr. Haimovich: An ECG (Fig. 2A) showed sinus rhythm; Mobitz type I second-degree atrioventricular block, along with first-degree atrioventricular block with a PR interval of 240 msec (reference range, 120 to 200); and Q waves in the inferior leads.

Diagnostic tests were performed, and management decisions were made.

Figure 1. Chest Radiograph.
A single-view frontal image obtained with a portable device shows no evidence of pneumonia or pulmonary edema and no pleural effusion. The heart size is normal. An implantable loop recorder projects over the left hemithorax. Electrocardiographic leads can also be seen.

DIFFERENTIAL DIAGNOSIS

Dr. Deborah Gomez Kwolek: This young woman who lives in a rural area began to have neck and arm pain and fatigue after recovering from infection with SARS-CoV-2. She was treated with a nonsteroidal antiinflammatory drug (NSAID) and a tapering dose of a glucocorticoid, presumably for cervical radiculopathy or a muscle strain, and her symptoms abated. However, she then began to have pleuritic and positional chest pain, palpitations and an irregular pulse, dyspnea on exertion, and myalgias, and the blood level of D-dimer was elevated. At the current presentation, the patient had Mobitz type I second-degree atrioventricular block and an elevated blood level of NT-proBNP. In working toward the most likely diagnosis, I will consider cardiac and pulmonary conditions that could account for these findings.

COMPLICATIONS OF INFECTION WITH SARS-COV-2

The patient's remote and recent infections with SARS-CoV-2 warrant evaluation for possible long-term consequences of coronavirus disease 2019 (Covid-19) such as carditis or multisystem inflammatory syndrome, a condition that is similar to Kawasaki's disease.[1,2] Multisystem inflammatory syndrome typically occurs 2 to 6 weeks after SARS-CoV-2 infection and is a consideration in this patient, particularly because she was not vaccinated against SARS-CoV-2. However, multisystem inflammatory syndrome is usually associated with elevated blood levels of inflammatory markers; this patient's erythrocyte sedimentation rate and C-reactive protein level were both normal. Pericarditis associated with Covid-19 may cause palpitations, and blood levels of inflammatory markers and troponin may be normal.[3-5] However, the most common arrhythmias in patients with pericarditis associated with Covid-19 are sinus tachycardia and, less commonly, repetitive supraventricular and ventricular premature beats; second-degree atrioventricular block is atypical in such patients.

PULMONARY EMBOLISM

Pulmonary embolism was appropriately considered in this patient who had presented with dyspnea on exertion, pleuritic chest pain, and an elevated blood level of D-dimer. Recent infection with SARS-CoV-2 is associated with thrombotic events in some patients. This patient's list of medications did not include oral contraceptives, but I would ask the patient specifically about any use of these medications, given that they can also contribute to thrombophilia. However, the results of contrast-enhanced CT angiography did not show evidence of pulmonary embolism, which effectively rules out this diagnosis.

ACUTE CORONARY SYNDROME

The development of atherosclerotic coronary vascular disease would be unusual in a person of this patient's age. However, this patient had a family history that may have increased her risk of heart disease, and she had a history of exposure to to-

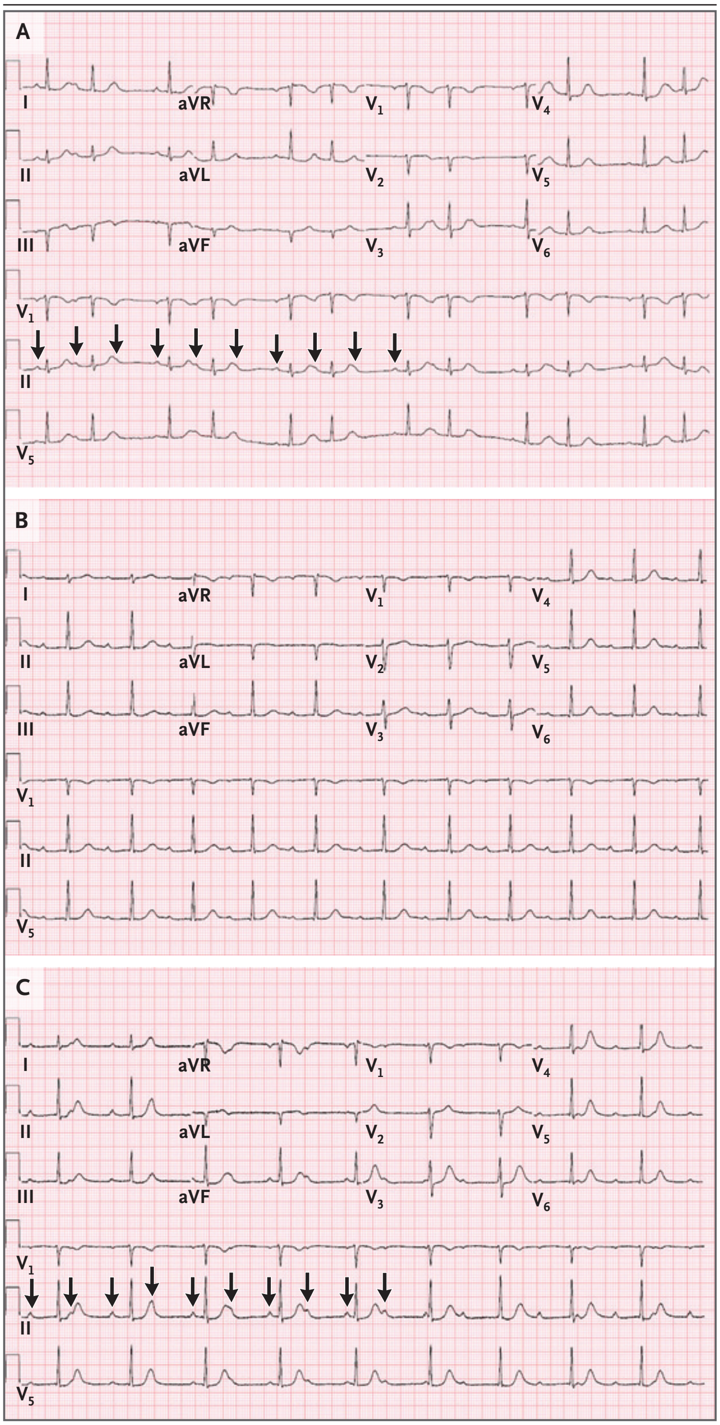

Figure 2. Electrocardiograms.

An electrocardiogram (ECG) obtained on presentation (Panel A) shows Mobitz type I second-degree atrioventricular block, along with first-degree atrioventricular block with a PR interval of 240 msec. The QRS complexes associated with Mobitz type I block appear in a regularly irregular — or "grouped beating" — cadence. An ECG obtained on hospital day 1 (Panel B) shows sinus rhythm with first-degree atrioventricular block with a PR interval of 350 msec. An ECG obtained on hospital day 2 (Panel C) shows sinus rhythm with complete atrioventricular block and junctional escape rhythm. Arrows in Panels A and C indicate P waves.

bacco, vaping, and marijuana. In addition, it is important to consider nonatherosclerotic causes of coronary disease, including spontaneous coronary-artery dissection or vasculitis. However, the positional and pleuritic aspects of the patient's chest discomfort are not typical of an acute coronary syndrome. Furthermore, ECG did not show ST-segment depression and the blood level of troponin was normal. Overall, an acute coronary syndrome is unlikely to explain this patient's presentation.

CARDIOMYOPATHY

Could the patient have cardiomyopathy leading to chest pain, dyspnea, and an elevated blood level of NT-proBNP? Treatment with hydroxychloroquine can cause toxic myopathy and cardiomyopathy.[6] I would ask the patient specifically about any undisclosed use of hydroxychloroquine for prophylaxis against, or treatment of, SARS-CoV-2 infection. Iron overload from ingestion of animal liver extract can also lead to cardiomyopathy; however, the blood level of ferritin was reportedly normal approximately 2 years before the current presentation. In addition, CT angiography of the chest did not show enlargement of the cardiac chamber, which makes the diagnosis of chronic cardiomyopathy unlikely.

INFECTION

The patient's nonspecific prodromal symptoms of myalgia and headache, followed by pleuritic chest pain and atrioventricular block, may be suggestive of an infectious process culminating in myocarditis. Although her recent treatment with an NSAID and a glucocorticoid could mask fever, the normal blood level of troponin and the absence of ST-segment changes on ECG make many infectious causes of myocarditis unlikely, with several exceptions, as discussed below.

Acute rheumatic fever and bacterial endocarditis with perivalvular or acute root abscesses can be associated with atrioventricular block and are important to rule out. Sheep and rabbits are kept on the patient's property; these animals are potential reservoirs for *Brucella melitensis* and *Francisella tularensis*.

Brucellosis can cause endocarditis and, in rare cases, can be associated with atrioventricular block.[7] Case reports have described tularemia endocarditis, but there is not a specific association with atrioventricular block.

LYME CARDITIS

Early consideration of Lyme carditis in a patient with atrioventricular block is extremely important, since conduction abnormalities can advance rapidly to third-degree atrioventricular block and may begin to abate within hours after the administration of antibiotic agents. This patient did not report having had a tick bite, but she lives in a wooded area of New England and spends time outdoors — factors that increase the likelihood that she had been exposed to ticks infected with *B. burgdorferi*, the causative agent of Lyme disease in the United States.

Lyme disease is associated with several clinical stages, and the stages may overlap.[8] Early localized disease occurs 3 to 30 days after inoculation and classically causes a rash (erythema migrans, the bull's-eye rash) at the site of a tick bite. The identification of erythema migrans is considered to be diagnostic of early Lyme disease, and when this rash is present, no further testing for Lyme disease is typically necessary. However, up to 40% of patients in whom Lyme disease is ultimately diagnosed are unaware of having been bitten by a tick and do not recall a rash consistent with erythema migrans. Patients with early localized Lyme disease may also have flulike symptoms such as fatigue, headache, and myalgias, which were reported by this patient; arthralgias (but not frank arthritis) are also common.

Early disseminated disease develops in up to 60% of untreated patients infected with *B. burgdorferi*. Days to weeks after infection, multiple erythema migrans lesions may appear on the skin. Weeks to months later, carditis, neurologic disease, or arthralgias and arthritis may develop in some patients. Lyme carditis is characterized by atrioventricular block and myopericarditis, which together could explain this patient's palpitations, irregular pulse, dizziness, dyspnea, and chest pain. Neurologic disease may be characterized by cranial neuropathy, lymphocytic meningitis, or painful radiculoneuropathy involving one or multiple dermatomes.

This patient's early symptoms of neck tightness and arm pain may have been musculoskeletal in nature but could also have been neurologic manifestations of early disseminated Lyme disease. Elevated levels of alanine aminotransferase and aspartate aminotransferase, as were seen in this patient, are well described in patients with Lyme disease. The patient had received glucocorticoids, which could have masked fever. Nevertheless, the absence of fever and the normal erythrocyte sedimentation rate and C-reactive protein level are also not inconsistent with this diagnosis.

To confirm the diagnosis of Lyme disease, I would perform serologic testing for antibodies against *B. burgdorferi*. While awaiting the results, I would treat the patient empirically for Lyme disease and would continue the use of mobile cardiac telemetry to monitor for worsened atrioventricular block.

CLINICAL IMPRESSION

Dr. Haimovich: Given the patient's cardiac conduction disease and the fact that she resided in the northeastern United States and had outdoor exposures, Lyme carditis was the leading diagnosis. Serologic testing for Lyme disease was performed, and empirical antibiotic therapy with intravenous ceftriaxone was initiated immediately. An ECG obtained on hospital day 1 (Fig. 2B) showed worsened prolongation of the PR interval, which had increased to 350 msec. An ECG obtained on hospital day 2 (Fig. 2C) showed complete atrioventricular block. The urgent placement of a temporary transvenous pacing wire was considered; however, the patient was asymptomatic and maintained cardiac output, and the complete atrioventricular block was transient and resolved within hours after onset.

The Patient: On my third day in the hospital, my friend (who is a nurse) reminded me to show Dr. Haimovich a picture I had taken with my phone (Fig. 3) of a rash I had 3 weeks before this hospitalization. The rash in the picture was on my arm, but at the time, I had similar rashes all over my body.

CLINICAL DIAGNOSIS

Lyme carditis.

Figure 3. Photograph of Previous Rash.
A photograph taken by the patient 3 weeks before the current presentation shows an erythematous, circular, macular rash on the left arm. Note that the small black mark inside the rash was present on the original photograph and is considered by the treating clinicians to be an artifact.

DR. DEBORAH GOMEZ KWOLEK'S DIAGNOSIS

Lyme disease, early disseminated stage with carditis.

DISCUSSION OF LABORATORY TESTING

Dr. Sarah E. Turbett: The diagnostic test performed was an enzyme immunoassay targeting *B. burgdorferi*–specific antibodies, which was positive. This result was confirmed by an immunoblot assay, which was positive for IgM and IgG. Microscopic examination of thick and thin blood smears for babesia were negative, as was a blood polymerase-chain-reaction assay for *Anaplasma phagocytophilum*.

Lyme disease (also known as Lyme borreliosis) is an infection caused by *B. burgdorferi*.[9] In the northeastern United States, transmission occurs through a bite from the *Ixodes scapularis* tick, and infection typically manifests in one of three stages.[9] Patients commonly have multiple stages of infection during the clinical course. Early localized disease (the first stage) occurs days to weeks after a tick bite and is characterized by erythema migrans,[9] as was described in this case. Early disseminated disease (the second stage) occurs weeks to months after a tick bite and is associated with systemic symptoms including fever, joint pain, and myalgias.[9] Neurologic and cardiac involvement can also occur.[9] Late disseminated disease (the third stage) commonly results in arthritis that affects large joints, with minimal systemic symptoms.[9]

The cornerstone of diagnosis of Lyme disease is serologic testing, which identifies *B. burgdorferi*–specific IgM and IgG.[10] Because the humoral response to infection develops over time, the sensitivity of serologic diagnosis of primary infection is low. At this stage, diagnosis is often made on the basis of clinical variables alone. In contrast, patients with secondary or tertiary infection have well-developed antibody responses. At these stages, serologic diagnosis is highly sensitive, with rates approaching 100% among patients with tertiary infection.[11]

Serologic testing for Lyme disease follows a two-tiered model, in which a highly sensitive test is followed by an orthogonal assay to confirm equivocal or positive results. The Centers for Disease Control and Prevention (CDC) endorses two strategies for two-tiered testing: standard two-tiered testing (STTT), in which an enzyme immunoassay is performed, followed by an immunoblot assay; and modified two-tiered testing (MTTT), in which an enzyme immunoassay is performed, followed by a second enzyme immunoassay with orthogonal properties (Fig. 4).[12] Both algorithms have good clinical performance in patients with secondary or tertiary infection, with advantages and limitations.

Advantages of STTT include its high specificity and its ability to provide information on the duration of infection, given that the immunoblot assay identifies many different *B. burgdorferi*–specific antibodies that develop at different time points in response to infection. The second test also allows clinicians to correlate antibody patterns with clinical symptoms, which can improve diagnostic accuracy in complex cases.[10] Limitations of STTT as compared with MTTT include its higher cost and complexity and its longer turnaround time.[10] The results of the immunoblot assay can also be somewhat subjective.

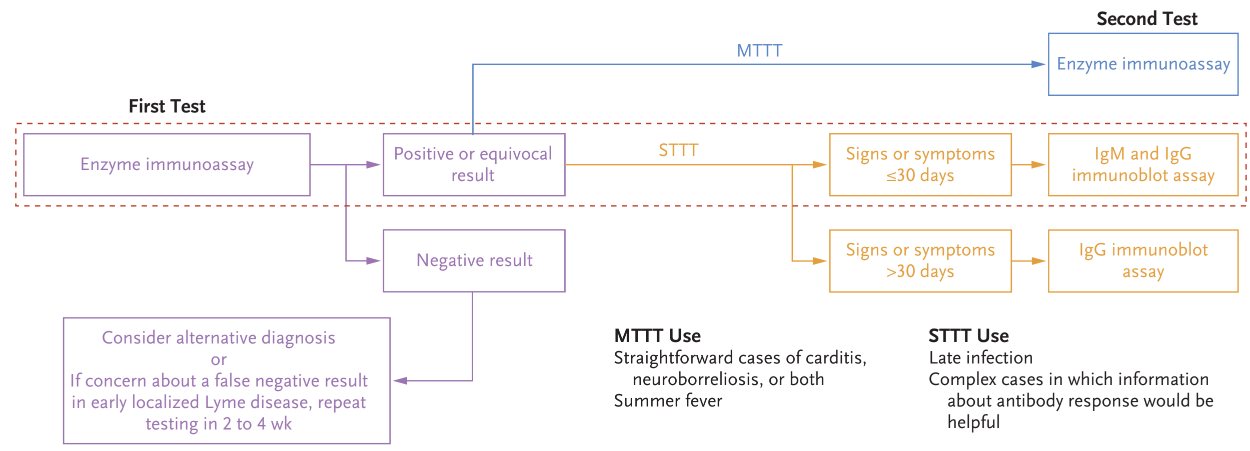

Figure 4. Two-Tiered Testing Algorithms for Diagnosis of Lyme Disease.

The steps in purple show components of the algorithm shared by both standard two-tiered testing (STTT) and modified two-tiered testing (MTTT). The steps in orange show components of the algorithm specific to STTT. The steps in blue show components of the algorithm specific to MTTT. The red dashed outline shows the STTT algorithm that was used for this patient, which included an enzyme immunoassay followed by an immunoblot assay; the patient had had signs or symptoms for 30 days or less when the algorithm was used to make the diagnosis. MTTT includes the use of only enzyme immunoassays that have been approved by the Food and Drug Administration. For STTT, an immunofluorescence assay can also be used as the first-tier test.

Advantages of MTTT over STTT include its higher sensitivity for the detection of primary infection, lower cost and complexity, and shorter turnaround time.[10] MTTT also provides results that are more objective than those provided by STTT. Despite its higher sensitivity among patients with primary infection, MTTT still has poor sensitivity at this stage of infection and should not be used to rule out acute infection if the clinical suspicion is moderate to high. MTTT also does not measure the antibody response to the extent that the immunoblot assay from the STTT algorithm does; therefore, results obtained with MTTT cannot be correlated with clinical symptoms to the same degree as those obtained with STTT.[10] Appropriate clinical scenarios for the use of STTT and MTTT are listed in Figure 4.

LABORATORY DIAGNOSIS

Infection with *Borrelia burgdorferi*.

DISCUSSION OF MANAGEMENT

Dr. David M. Dudzinski: Lyme carditis is a clinical manifestation of Lyme disease in approximately 1% of cases in the United States, and it is a less common manifestation than erythema migrans, arthritis, facial palsy, or encephalitis. This statistic is based on a stringent CDC definition of Lyme carditis as sudden-onset but transient second- or third-degree atrioventricular block.[13] Lyme carditis is more common in men than in women, and the strongest predictor of third-degree atrioventricular block is a PR interval of greater than 300 msec, which this patient had. Other types of arrhythmia such as sinoatrial exit block or bundle-branch block have been reported in patients with Lyme carditis but are uncommon[13]; no other arrhythmia was recorded in this patient during her hospitalization. Certain symptoms of Lyme carditis can be attributed to bradycardia, such as fatigue, dyspnea, and intermittent palpitations, which were seen in this patient; other such symptoms may include chest discomfort, lightheadedness, and syncope.

Atrioventricular block can fluctuate as it did in this patient, but third-degree atrioventricular block in particular is typically transient (median time to resolution, 6 days). A temporary transvenous pacing wire should be considered for standard indications including advanced atrioventricular block with hemodynamic consequences and a refractory response to medical therapies.[14] Because the conduction disturbance is almost always reversible with antibiotic treatment, implantation of a permanent pacemaker is usually not needed.[13,14]

Dr. Turbett: Antimicrobial therapy for Lyme carditis follows recommendations outlined by members of the Infectious Diseases Society of America, the American Academy of Neurology, and the American College of Rheumatology and is based on the severity of illness. Patients with mild-to-moderate cardiac manifestations and no risk factors for severe complications can be treated in the outpatient setting with oral antibiotics. Oral antibiotic options include doxycycline, amoxicillin, cefuroxime axetil, and azithromycin. Patients with severe cardiac manifestations or risk factors for other severe complications typically receive intravenous ceftriaxone. Once clinical improvement (resolution of arrythmia or a decrease in the PR interval to below 300 msec) occurs, patients can be transitioned to one of the oral therapies mentioned above. The recommended duration of treatment is 14 to 21 days.[15]

FOLLOW-UP

Dr. Haimovich: The patient's PR interval decreased, and she was discharged home on hospital day 4 with instructions to complete a 3-week course of oral doxycycline. The patient was seen in the cardiology clinic 2 weeks after discharge. She continued to have fatigue but had resumed participation in low-intensity exercise. The first-degree atrioventricular delay that had been seen on ECG resolved within 2 weeks after onset. During the next few months, her functional status slowly improved, and she was able to return to her previous activity level.

PATIENT PERSPECTIVE

The Patient: I found out later that I had had intermittent complete heart block when I was evaluated at the second hospital, but I do not know why this was not acted on. I had explained that I am a fit woman and that I knew something was wrong, but I was told that I was anxious. I even suggested a Lyme test but was told it was not useful. When I presented to this hospital, I felt heard, and I did not feel judged for my beliefs in holistic and alternative medicine practice.

It was 4 months until I felt that I was back to normal activity. Emotional recovery was also part of my recuperation from Lyme carditis. Being here today, and being back to being healthy, is gratifying and provides some closure. I want other clinicians to learn from my experience and to always really listen to the patient.

ADDITIONAL DISCUSSION

Dr. Dudzinski: Do the specific bands seen on an immunoblot assay correlate with different geographic regions or microbiologic features or perhaps different manifestations of Lyme disease (e.g., carditis or neuritis)?

Dr. Turbett: The prime correlation is with the stage of the infection, since certain bands are observed early as opposed to late. This is why a certain number of *B. burgdorferi*–specific protein bands are needed to confirm the diagnosis. However, borrelia species found in Europe will have a pattern of protein bands that is different from that seen in borrelia species found in the northeastern and midwestern United States.

FINAL DIAGNOSIS

Lyme carditis.

This case was presented at the Medicine Case Conference Interview Series.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank the patient for attending the conference and sharing her perspective and experience.

AUTHOR INFORMATION

[1] Department of Medicine, Massachusetts General Hospital, Boston; [2] Department of Medicine, Harvard Medical School, Boston; [3] Department of Radiology, Massachusetts General Hospital, Boston; [4] Department of Radiology, Harvard Medical School, Boston; [5] Department of Pathology, Massachusetts General Hospital, Boston; [6] Department of Pathology, Harvard Medical School, Boston.

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